SOFTWARE ARCHITECTURE

Escuela de Ingeniería Informática

Universidad de Oviedo

Lab 6

TDD: Test-Driven Development
Code coverage (SonarCloud)
Continuous integration (GitHub Actions)
Static analysis tools (SonarCloud)

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TDD

Software development process where requirements are converted to specific test cases

The opposite to software development that allows non-tested software to be deployed

Technique proposed by Kent Beck

TDD - Phases

1. Add a test case
2. Execute test cases
   - new one fails
3. Write the code
4. Execute all test cases
5. Code refactor

TDD - Features

Simple code created to satisfy the test case
We get clean code as a result
And a test-suite
Helps focus to know what we want to implement

Code Coverage (SonarCloud)

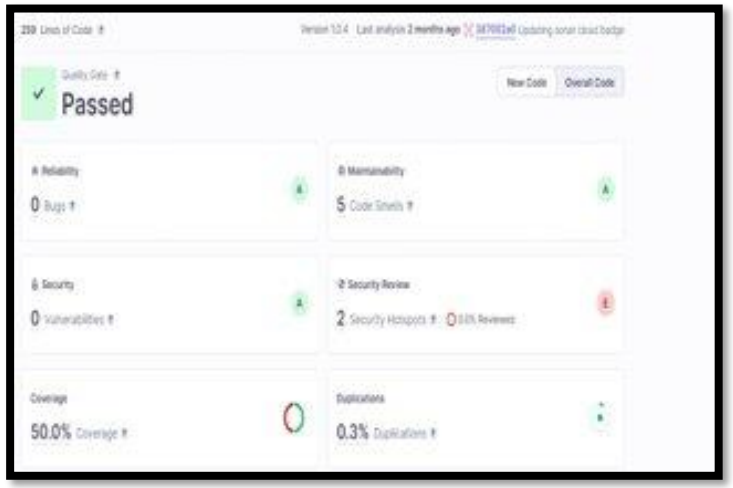
- Code coverage: Measure to show what code lines has been executed by a test suite
- Tool that includes code coverage as a metric in the code evaluation process
- Some terminology about SonarCloud:
  - LC: lines_to_cover – uncovered_lines
  - EL: lines_to_cover

Code Coverage in SonarCloud

- Coverage ratio is calculated with the formula:

  LC/EL

- After the tests, it generates a file that allows to do the analysis
  - https://sonarcloud.io/summary/overall?id=Arquisoft_yovi_???



TDD – Example test

- Testing a basic component in React.js (App.test.js)

```
import { render, screen } from '@testing-library/react';
import App from './App';

test('renders welcome message', () => {
  render(<App />);
  const welcomeMessage = screen.getByText(/Welcome to the 2025 edition of the Software Architecture course/i);
  expect(welcomeMessage).toBeInTheDocument();
});
```

TDD – Example test

- Checking that the AddUser component works well:
  - Sometimes we need to mock some part of the application
  - If we didn't mock the api, our test would depend on the *userservice*
  - As these are unitary tests, we simulate that part of the app

```
14  it('should add user successfully', async () => {
15    render(<AddUser />);
16
17    const usernameInput = screen.getByLabelText(/Username/i);
18    const passwordInput = screen.getByLabelText(/Password/i);
19    const addUserButton = screen.getByRole('button', { name: /Add User/i });
20
21    // Mock the axios.post request to simulate a successful response
22    mockAxios.onPost('http://localhost:8000/adduser').reply(200);
23
24    // Simulate user input
25    fireEvent.change(usernameInput, { target: { value: 'testUser' } });
26    fireEvent.change(passwordInput, { target: { value: 'testPassword' } });
27
28    // Trigger the add user button click
29    fireEvent.click(addUserButton);
30
31    // Wait for the Snackbar to be open
32    await waitFor(() => {
33      expect(screen.getByText(/User added successfully/i)).toBeInTheDocument();
34    });
35  });
```

Continuous Integration (CI)

- Development practice that promotes developers to **integrate** code into a shared repository several times a day
- Every task to build the software is executed when some condition is met
  - For instance, push, pull request, or the creation of a new release

Continuous Integration (CI)

- Detect and solve problems continuously
- Always available
- Immediate execution of unit test cases and E2E tests.
- Automatic deployment
- Project quality monitorization.

Continuous Integration (CI)

- Examples:
  - Jenkins
  - Pipeline
  - Hudson
  - Apache Continuun
  - Travis
  - **GitHub Actions**

Continuous Integration (CI) - Uses

- Common usages:
  - Maintenance of the code in a repository
  - Building automation
  - Quick building
  - Execute test cases in a cloned production environment
  - Show results of last build.

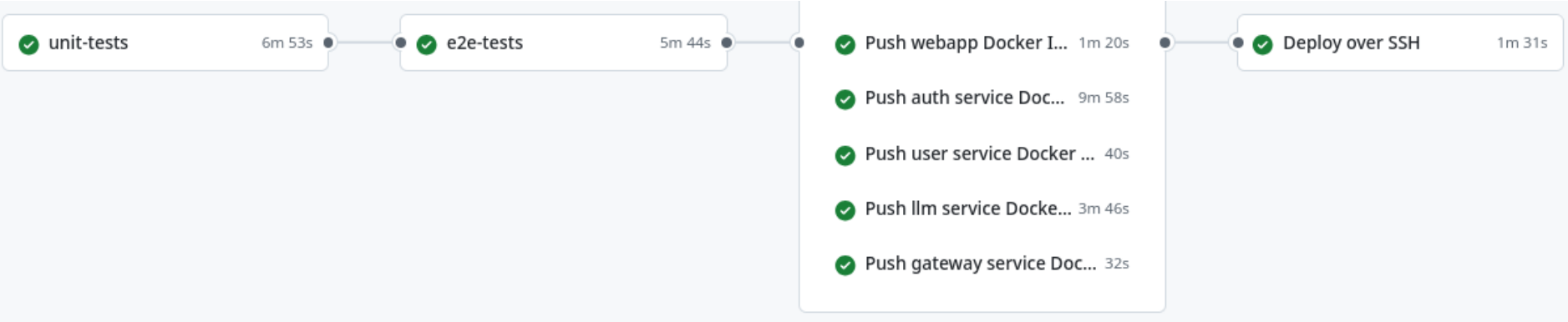
GitHub Actions

- Continuous integration service for projects stored in GitHub
- Free for free software projects
- Configuration is in one or multiple YAML files inside:
  `.github/workflows`

GitHub Actions

- .yml specifies:
  - Conditions for firing the process
  - List of jobs
    - Each executed in a specific environment
  - Steps to carry out the job (checkout, install dependencies, build and test)

```yaml
jobs:
  unit-tests:
    runs-on: ubuntu-latest
    steps:
    - uses: actions/checkout@v4
    - uses: actions/setup-node@v4
      with:
        node-version: 22
    - run: npm --prefix users/authservice ci
    - run: npm --prefix users/userservice ci
    - run: npm --prefix llmservice ci
    - run: npm --prefix gatewayservice ci
    - run: npm --prefix webapp ci
    - run: npm --prefix users/authservice test -- --coverage
    - run: npm --prefix users/userservice test -- --coverage
    - run: npm --prefix llmservice test -- --coverage
    - run: npm --prefix gatewayservice test -- --coverage
    - run: npm --prefix webapp test -- --coverage
    - name: Analyze with SonarQube
      uses: SonarSource/sonarqube-scan-action@master
      env:
        SONAR_TOKEN: ${{ secrets.SONAR_TOKEN }}
```



GitHub Actions

- Each job can have a specific purpose
  - Test a part of the app, deploy, etc.
- GitHub actions can be used to automate other parts of the repository.
  - Example: autoreply to new issues created in the repository

GitHub Actions

- We have jobs also to build the docker images and publish them to github
- Check the full documentation for the CI configuration

```yaml
docker-push-webapp:
  name: Push webapp Docker Image to GitHub Packages
  runs-on: ubuntu-latest
  permissions:
    contents: read
    packages: write
  needs: [e2e-tests]
  steps:
  - uses: actions/checkout@v4
  - name: Publish to Registry
    uses: elgohr/Publish-Docker-Github-Action@v5
    env:
      API_URI: http://${{ secrets.DEPLOY_HOST }}:8000
    with:
        name: arquisoft/wichat_0/webapp
        username: ${{ github.actor }}
        password: ${{ secrets.GITHUB_TOKEN }}
        registry: ghcr.io
        workdir: webapp
        buildargs: API_URI
```

Static analysis of the code

Analyze the code without compiling it based in rules
Detects bugs, code smells, system vulnerabilities, etc.
Useful to control the code quality.
If the code does not meet the quality requirements, then the commit can be blocked

Static Analysis - SonarCloud

sonarcloud

Static code analysis tool

It needs:

- Git server like GitHub
- Repository access
- An accepted language

Two types of analysis configuration:

- **Automated Analysis** (Default). Code coverage not available. Scanner running in SonarCloud servers
- **CI-based analysis**. Sonar scanner running at the project server and sending reports to SonarCloud.

Sonarlint

SonarLint detects and highlights issues that can lead to bugs, vulnerabilities, and code smells in your IDE (available in the popular ones e.g. IntelliJ, Visual Code, Visal Studio, Eclipse…)

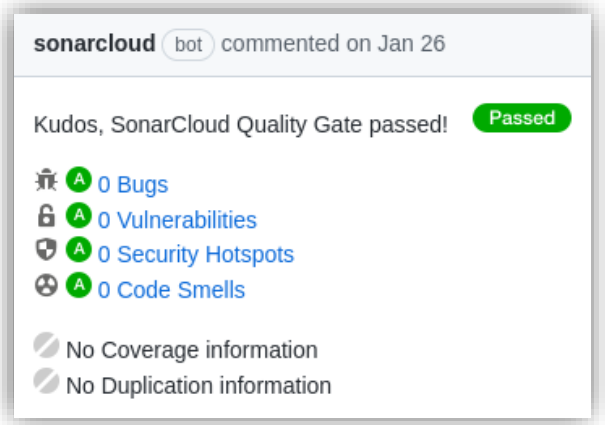
The análisis is performed locally (before the changes are submitted to the repository), can be executed:

- Manually
- Automatically over the changed files before the commit-push.

For further details regarding supported IDEs, languages and installation instructions, please visit the oficial webpage

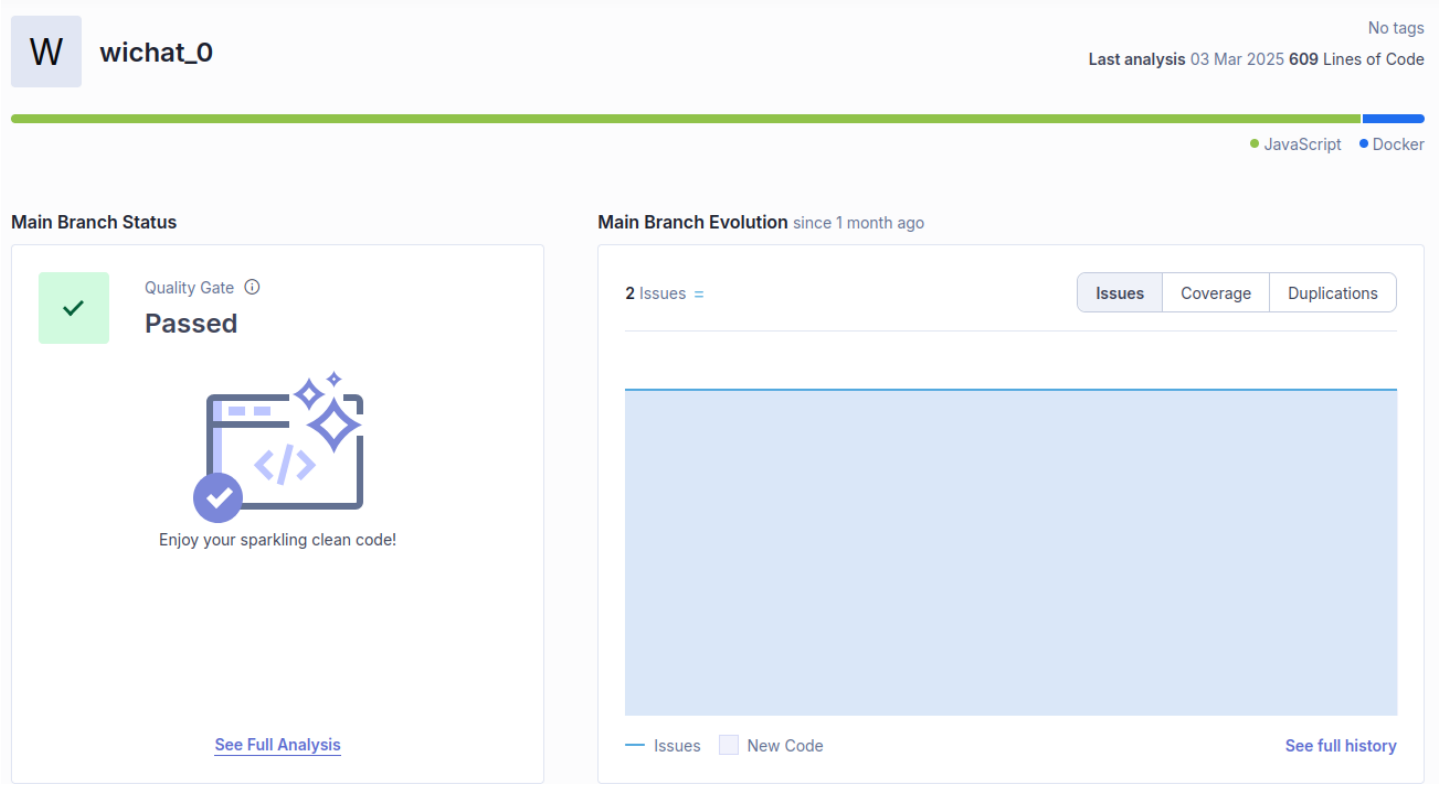
SonarCloud – yovi_x configuration

After changes are pushed to the repository (example, a new pull request) We have information about the code quality of the pull request that we are merging to our project

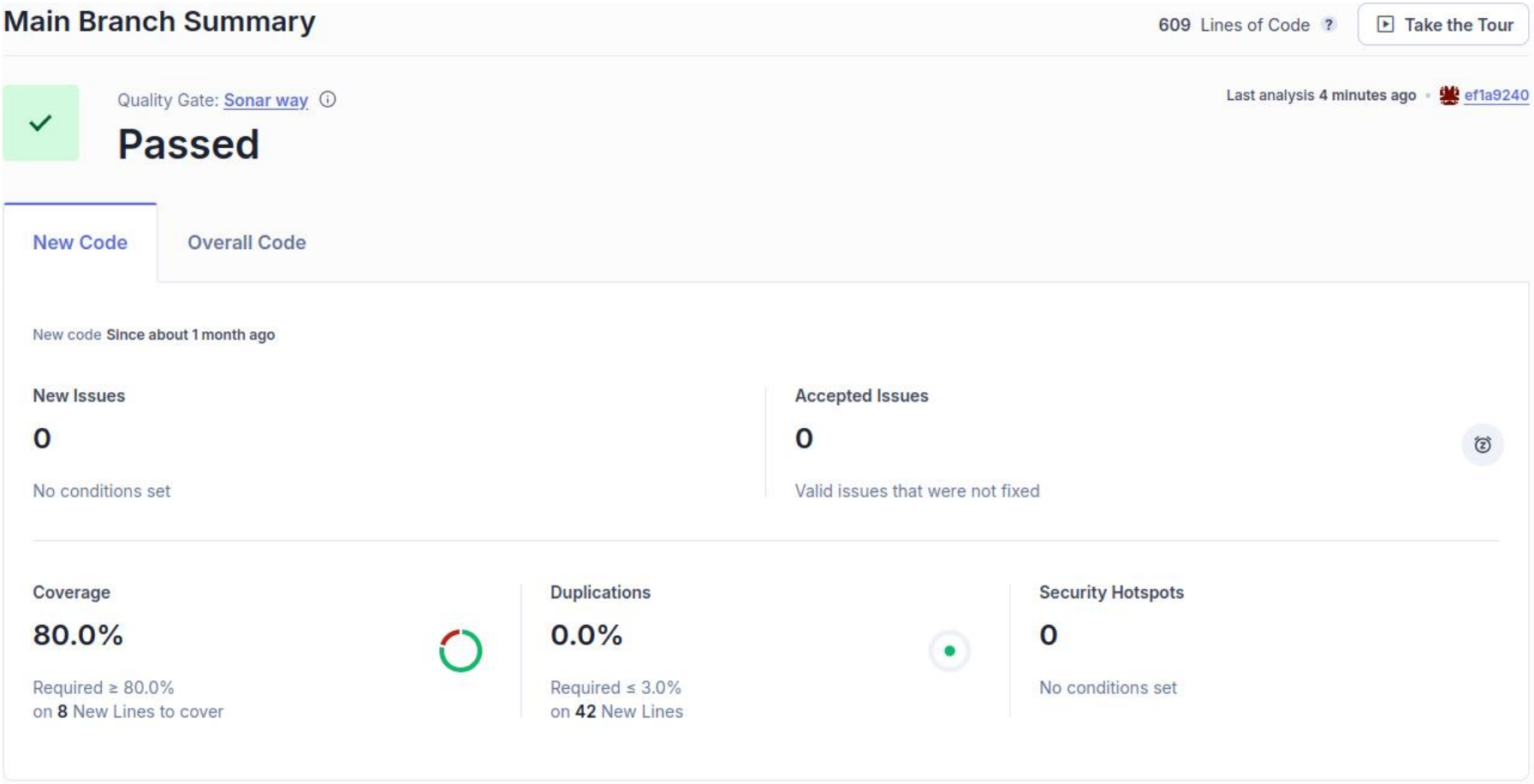


SonarCloud

In the Project Dashboard we can check project last analysis in the main branch, pull request and specific branches

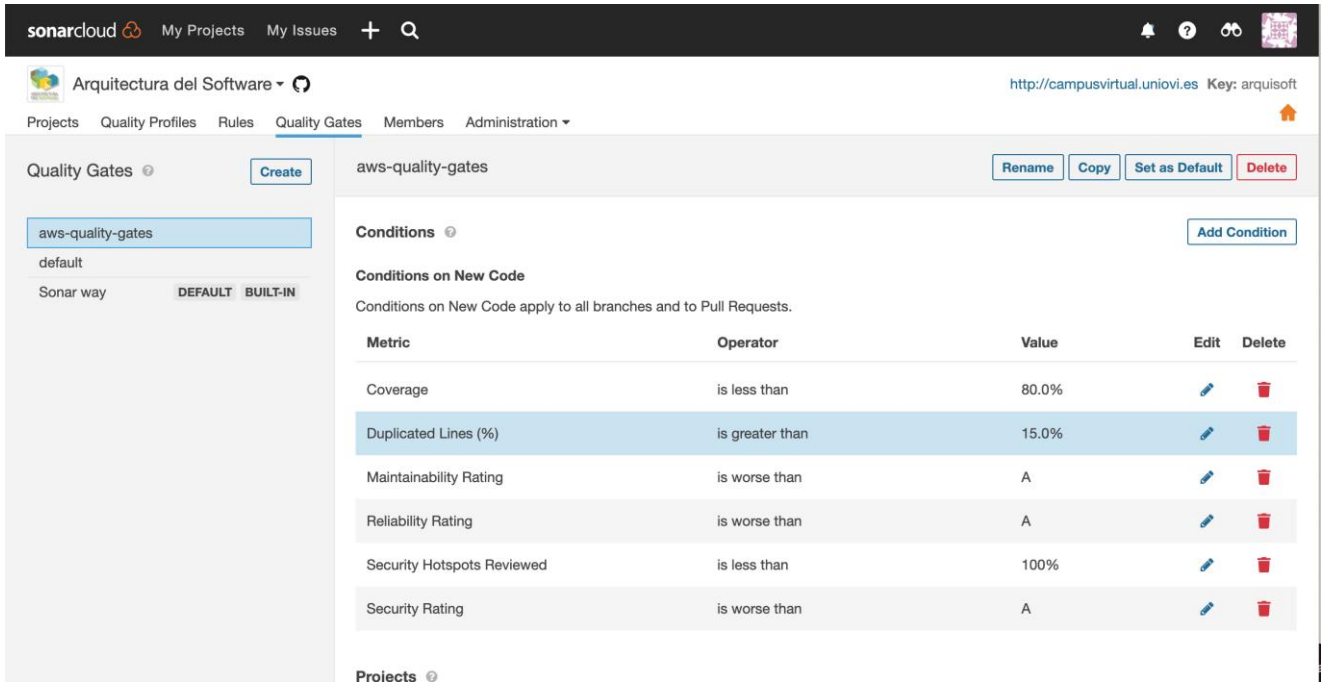


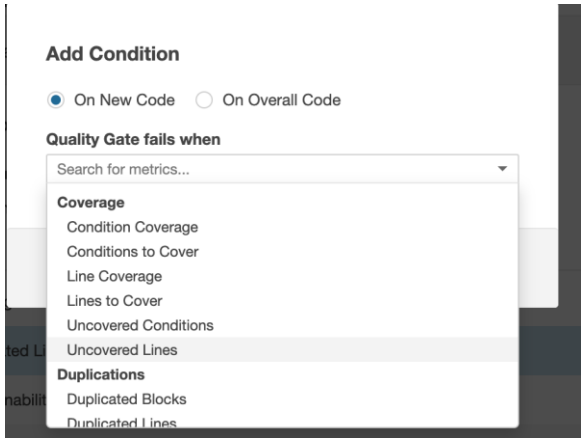
SonarCloud: Project certification and Quality evolution



SonarCloud: Quality Gates

At organization level, we can define the Quality Gates that our project must pass.





Example AWS-Quality-Gates , we increase the procentage of duplicate lines that can be found before launch exception

SonarCloud: Quality gates

A **Quality Gate** is a set of conditions that our project should meet.

That conditions include different aspect: code coverage, static code analyse based in rules, code duplicated, ..

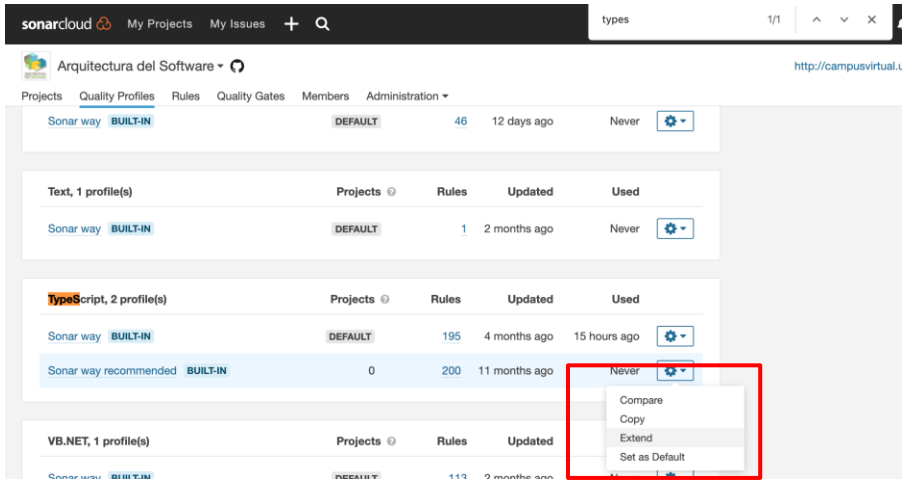
yovi_o default project uses code coverage with SonarCloud

SonarCloud: Profiles and Rules

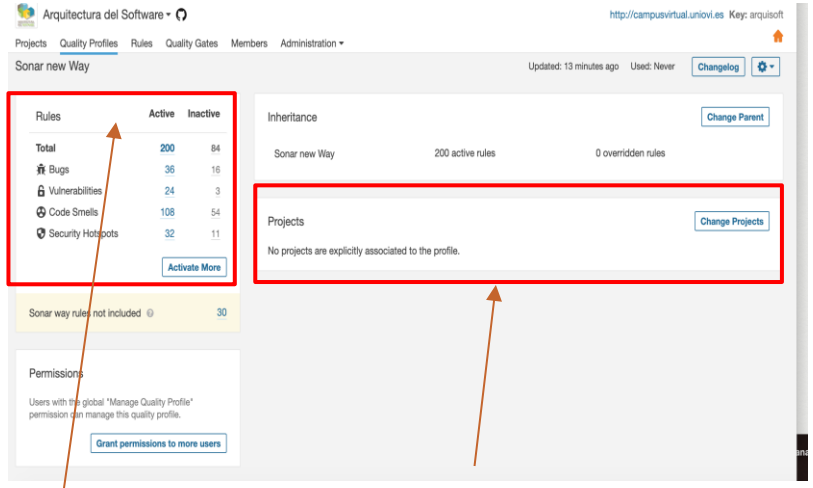
Rules are defined at profile level

We can add, desactivate, update rules creating a new profile :

Copy a parent profile - change it - associate it to the project



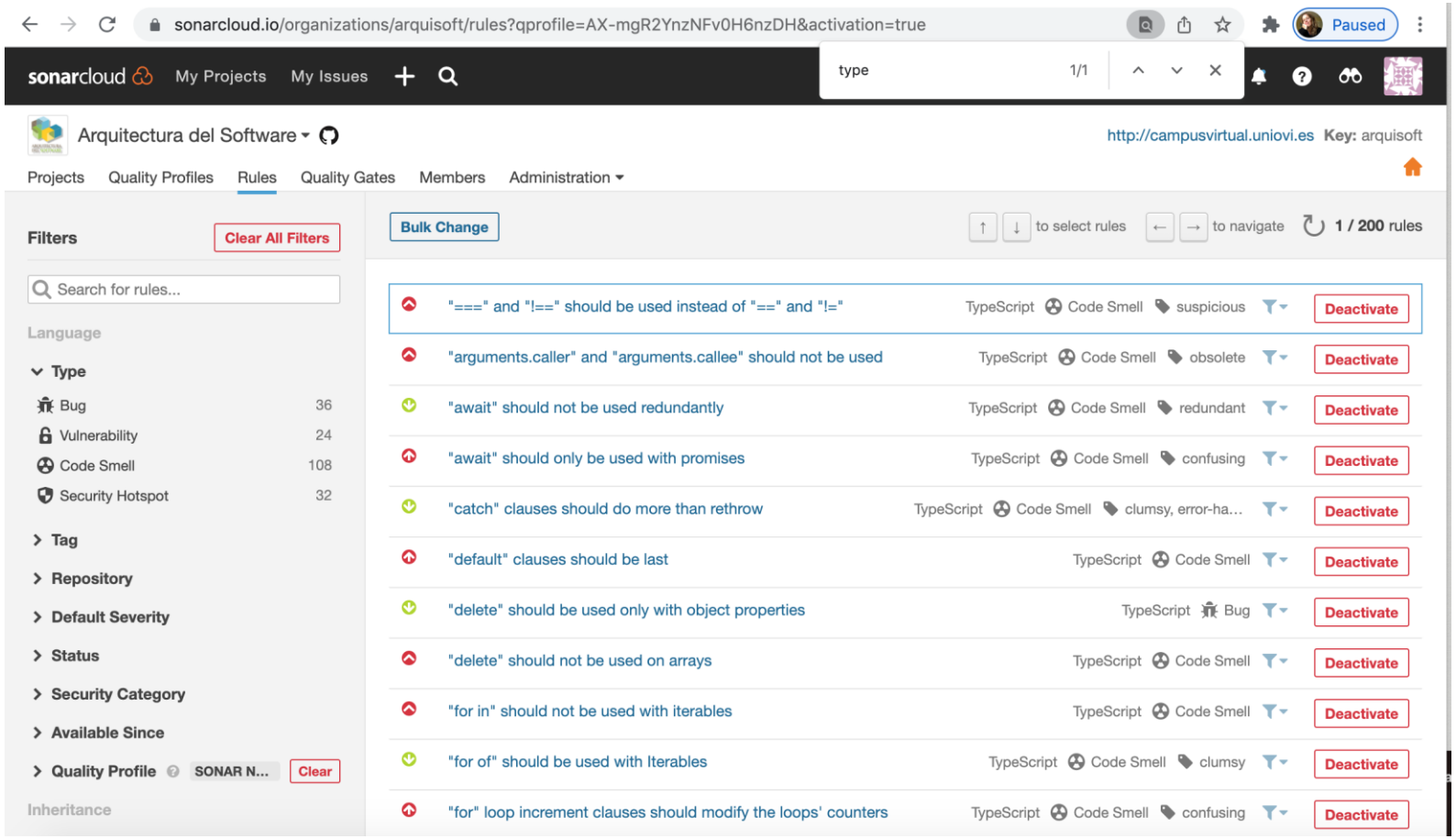
Create a new profile

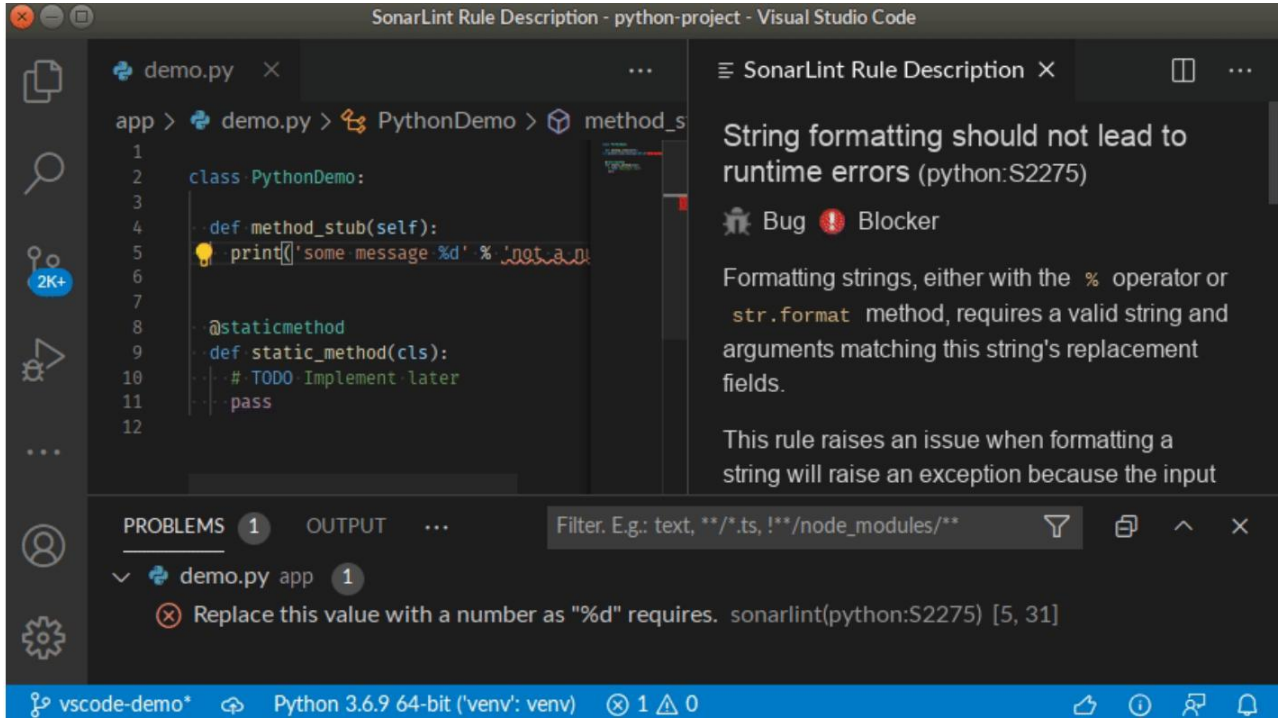


Set the profile rules

Associate the profile to the project

Rules configuration



View alerts when coding

- https://marketplace.visualstudio.com/items?itemName=SonarSource.sonarlint-vscode